

ETF Rotation Backtest: Version 2

Weekly ETF rotation on QuantConnect (LEAN, Python). Development period 2008 to 2023, holdout 2024 to 10 July 2026.

Bottom line. Version 2 cut drawdown compared with SPY, but it did not beat SPY or a plain 60/40 portfolio on the holdout period. In the development period it earned 9.6% a year with a 22.7% max drawdown (SPY: 9.8% a year, 51.9% drawdown). In the holdout, run once, it earned 10.8% a year with a Sharpe of 0.55, against 22.2% and 1.06 for SPY and 14.3% and 0.96 for the 60/40. The extra risk switches I tested did not earn their place, so the final version has all of them off.

1. Rules

Universe of 12 ETFs (SPY, QQQ, IWM, EFA, EEM, XLF, XLE, VNQ, TLT, IEF, GLD, DBC), with BIL as cash and HYG used only as a signal. Each week, on the first trading session, the decision is made on the close and traded at the next open. Score = $(0.4 \times 21 \text{ day return} + 0.4 \times 63 \text{ day return} + 0.2 \times 126 \text{ day return})$ divided by annualized 20 day volatility. An ETF is eligible if its price is above its 100 day average and its score is above zero. Hold the top 3 eligible, and keep an existing holding while it stays eligible and in the top 5. Empty slots go to BIL. Weights are inverse 20 day volatility, capped at 40% each, with the excess redistributed and the remainder in BIL. Weight changes under 2% are skipped, and full exits always execute. Costs: Interactive Brokers fee model plus 2 basis points of slippage per side.

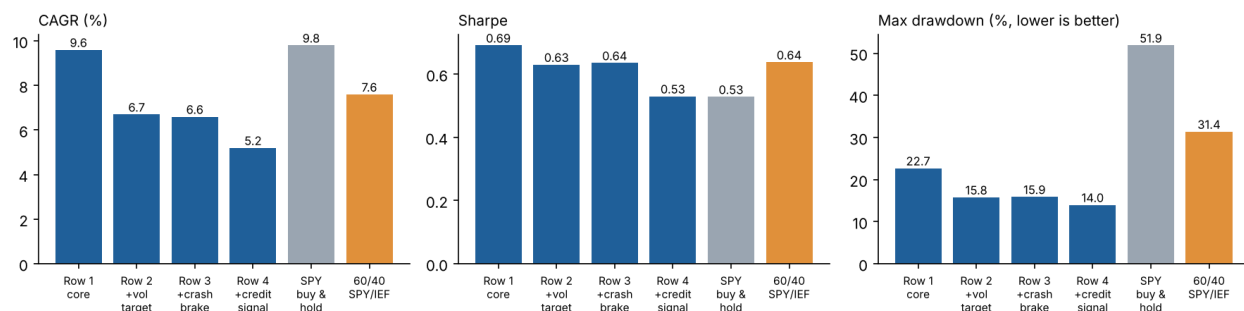
Three optional switches were tested: a volatility target (10%), a crash brake (SPY more than 5% below its 20 day high caps exposure at 50%), and a credit signal (HYG/IEF below its 50 day average caps exposure at 50%). If several caps apply, the lowest is used.

2. Ablation, development period (2008 to 2023)

	CAGR	Vol	Sharpe	Max DD	Worst year	Underwater	Turn/yr	Trades
Row 1: core only (chosen)	9.6%	13.5%	0.6912	-22.7%	-10.9% (2015)	1,009 days	1,624%	1,345
Row 2: + vol target	6.7%	10.0%	0.63	-15.8%	-10.0% (2015)	1,009 days	1,863%	2,374
Row 3: + crash brake	6.6%	9.7%	0.6353	-15.9%	-10.6% (2015)	950 days	2,241%	2,667
Row 4: + credit signal	5.2%	8.7%	0.53	-14.0%	-7.5% (2015)	1,287 days	2,834%	2,917
SPY buy and hold	9.8%	20.1%	0.5283	-51.9%	-36.2% (2008)	1,210 days	6%	1
60/40 SPY/IEF	7.6%	11.2%	0.6377	-31.4%	-17.5% (2008)	854 days	32%	384

Rows 2 and 4 show Sharpe to two decimals because that is what was read from the log before the printout was changed to four decimals.

Ablation on the development period (2008 to 2023)



Row 2 and Row 4 Sharpe are rounded to 2 decimals in the log. Row 1, Row 3 and the benchmarks to 4.

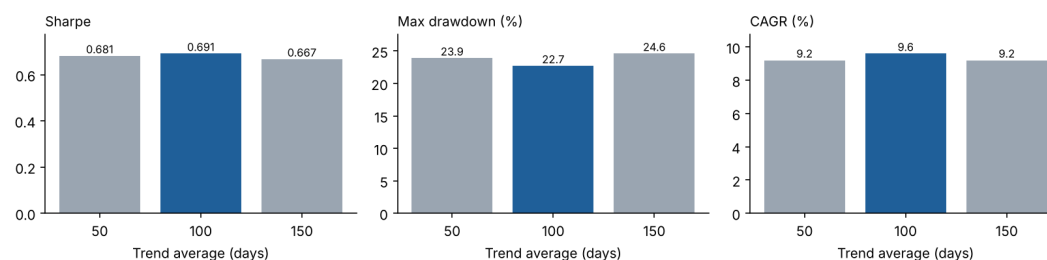
3. How the final version was picked

The rule was written before any results: pick the simplest row whose Sharpe is within 0.05 of the best row, and among those the one with the lowest max drawdown. Row 1 has the best Sharpe (0.6912). Row 3 was the close call at 0.6353, which is 0.0559 lower and just outside the line, so it is out. Rows 2 and 4 are further away. Only Row 1 remains. The switches cut drawdown, but they also cut the return by roughly a third or more and raised turnover, and none of them improved Sharpe.

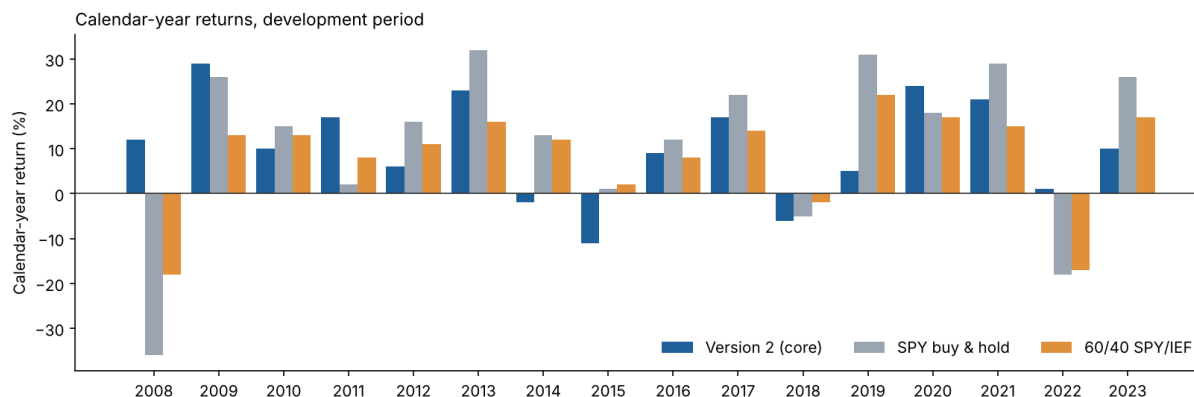
4. Sensitivity of the trend average (development period)

	CAGR	Vol	Sharpe	Max DD	Worst year	Underwater	Turn/yr	Trades
50 days	9.2%	13.1%	0.6813	-23.9%	-12.6% (2015)	770 days	2,084%	1,508
100 days (chosen)	9.6%	13.5%	0.6912	-22.7%	-10.9% (2015)	1,009 days	1,624%	1,345
150 days	9.2%	13.4%	0.6671	-24.6%	-10.6% (2015)	1,107 days	1,569%	1,331

Sensitivity to the trend average, development period (bars start at zero)



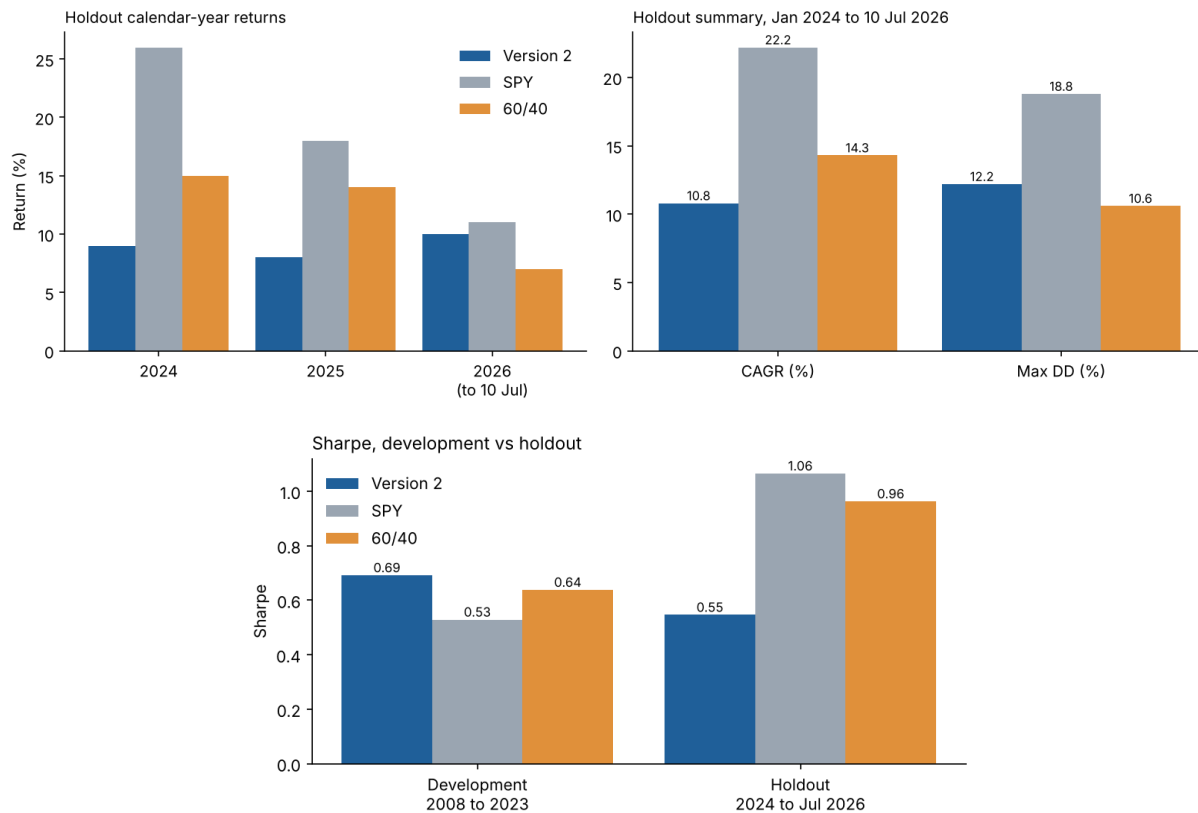
The results do not swing much: Sharpe stays between 0.67 and 0.69, and there is no sharp peak at 100 days. Time underwater moves more, but that comes from a few specific dips. The vol target and crash brake are off in the chosen version, so only the trend average was varied. The 50 day run had two orders rejected for insufficient buying power (2012-08-14 and 2013-06-24), so that row is slightly less clean.



5. Holdout, run once (Jan 2024 to 10 Jul 2026)

	CAGR	Vol	Sharpe	Max DD	Underwater	Turn/yr	Trades
Version 2	10.8%	12.2%	0.5459	-12.2%	393 days	1,565%	242
SPY buy and hold	22.2%	15.9%	1.0648	-18.8%	127 days	40%	1
60/40 SPY/IEF	14.3%	9.9%	0.9625	-10.6%	186 days	58%	62

Version 2 ended at \$129,398.52 from \$100,000 (+29.4%) with \$422.76 in fees (QuantConnect figures). Calendar years: Version 2 +9%, +8%, +10% (2026 to date); SPY +26%, +18%, +11%; 60/40 +15%, +14%, +7%.



Reading. The strategy was steady and had a smaller drawdown than SPY, but it gave up a lot of return in a strong stock market and lost to the 60/40 on return per unit of risk. Sharpe fell from 0.69 in development to 0.55 in the holdout. Two and a half years, one run, cannot say much either way, but it gives no evidence that this strategy adds value over simple alternatives.

QuantConnect's own equity chart for the holdout run:



6. Limitations

(1) One short holdout of about 2.5 years, run once. (2) Turnover is very high (about 1,600% a year in the core version) and is not yet explained; dollar fees are small in this test (\$6,680 over 16 years on \$100,000), but the number looks too high for a weekly three-holding rotation. (3) The 50 day sensitivity run had two orders rejected for insufficient buying power; I think it is an order-sizing issue around full exits and re-entries and it is not fixed. (4) The SPY and 60/40 benchmarks are simulated inside the algorithm with the same 2 bps cost. (5) Two ablation rows show Sharpe to two decimals only. (6) Free-tier QuantConnect data ending 10 July 2026, with results read from the algorithm's own log table, not an exported equity curve. (7) No like-for-like comparison with Version 1, which started in March 2010 and used other rules.

7. What I got wrong in Version 1

Version 1 reported a single backtest over the whole period (March 2010 to July 2026) that I had looked at while building the rules, so the numbers were likely flattering, and there was no real out-of-sample test. It did not beat SPY (about \$376,259 against about \$913,748 for SPY buy and hold; 8.43% a year, 17.6% max drawdown, Sharpe 0.458 on QuantConnect's calculation), and I should have led with that. It stacked several rules without testing whether each one helped, it had no rule for choosing the final version, and its own limitations list one invalid BIL order. Version 2 fixes the testing process (ablation, development and holdout periods, a selection rule written before the results). It is not a winner. It is a cleaner and more honest test.